

Lab Goal : This lab was designed to teach you more about a linked list and passing a linked list as a parameter. In this lab there is no `LinkedList` object. The list consists of a bunch of `ListNode` objects that you link together in your main method and then pass to the methods you write in the `ListFunHouse` class.

Lab Description : Use `ListNode` to write some basic `LinkedList` methods.

PART 1 – Open the `ListFunHouse.java` fill and complete the methods in this class.

PART 2 – Use `lab15b.java` to test your `ListFunHouse` class.

ListNode – stores a value and a reference to the next node

```
public class ListNode implements Linkable
{
    private Comparable listNodeValue;
    private ListNode nextListNode;

    public ListNode(Comparable value, ListNode next){
        listNodeValue=value;
        nextListNode=next;
    }

    public Comparable getValue(){
        return listNodeValue;
    }

    public ListNode getNext(){
        return nextListNode;
    }

    public void setValue(Comparable value){
        listNodeValue = value;
    }

    public void setNext(Linkable next){
        nextListNode = (ListNode)next;
    }
}
```

Sample Data :

See the main of `ListFun_Static_Runner`. There are also separate runners for each of the methods you'll write.

Sample Output :

```
Original list values:  go on at 34 2.1 -a-2-1 up over
num nodes = 8
List values after calling nodeCount:  go on at 34 2.1 -a-2-1 up over
List values after calling doubleFirst:  go go on at 34 2.1 -a-2-1 up over
List values after calling doubleLast:  go go on at 34 2.1 -a-2-1 up over over
List values after calling skipEveryOther:  go on 34 -a-2-1 over
List values after calling removeXthNode(2):  go 34 over
List values after calling setXthNode(2,one):  go one over
```